MIKE: fast and compact post-quantum NIKE

Max Duparc, on behalf of the MIKE team
LASEC, EPFL

Isogeny Club Season 9: November 10, 2026

The MIKE team

- Andrea Basso
- Pierrick Dartois
- Max Duparc
- Jonathan Komada Eriksen
- Sabrina Kunzweiler
- Michael Meyer
- Giacomo Pope
- Krijn Reijnders
- Damien Robert
- Ryan Rueger
- Sina Schaeffler

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Figure: The MIKE team in a nutshell

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A short history of Diffie-Hellman

DH Key Exchange

Alice		**Bob**
$a \in_\$ \mathbb{Z}_q$		$b \in_\$ \mathbb{Z}_q$
$A := g^a$		$B := g^b$
	$A \longrightarrow$	
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Less structure $\implies$ More security

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MIKE is a NIKE based on a symmetric monoid action.

MIKE (in practice)

Let $p = 2^f c - 1$, $E_0 : y^2 = x^3 - x$ and $\langle P_0, Q_0 \rangle = E_0[2^f]$

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- Let $E_A[2^f] = \langle P, Q \rangle$, $x_B \in \mathbb{Z}_{2^f}$ odd

$$F_B : (E_A \times {E_A}^{(p)})^2 \longrightarrow X_{AB}$$

$$\ker(F_B) := \begin{cases} T_1 & := \Big([x_B(1-x_B)/2]P, [x_B(1-x_B)/2]\pi(P), [x_B(1+x_B)/2]P, [x_B(1+x_B)/2]\pi(P)\Big) \\ T_2 & := \Big([x_B(1-x_B)/2]Q, [x_B(1-x_B)/2]\pi(Q), [x_B(1+x_B)/2]Q, [x_B(1+x_B)/2]\pi(Q)\Big) \\ T_3 & := \Big([(x_B+1)/2]Q, -[(x_B+1)/2]\pi(Q), [(x_B-1)/2]Q, -[(x_B-1)/2]\pi(Q)\Big) \\ T_4 & := \Big(-[(x_B+1)/2]P, [(x_B+1)/2]\pi(P), -[(x_B-1)/2]P, +[(x_B-1)/2]\pi(P)\Big) \end{cases}$$

MIKE (in practice)

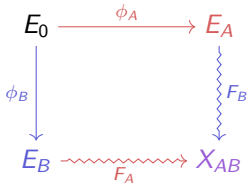
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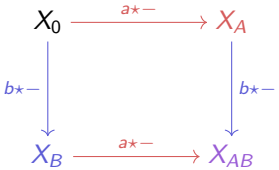


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Trivially

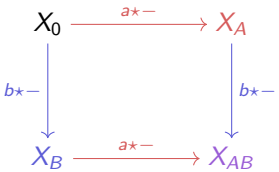
For all $x_A, x_B \in \mathbb{Z}_{2^f}$ odd, above diagram is commutative.

Today's presentation

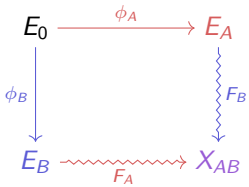


Monoid action side

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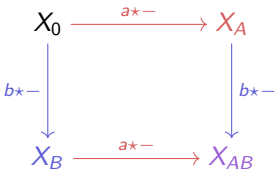


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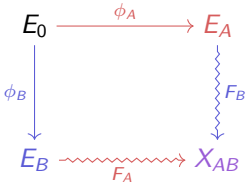
Isogeny side

Today's presentation



Monoid action side
(great for proves)

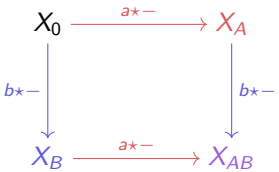




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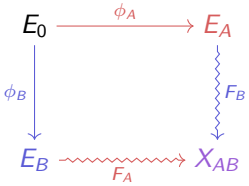
Journey from MIKE [Rob24] to MIKE [BDD+26]

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Journey from MIKE [Rob24] to MIKE [BDD+26]

- Lots of cool maths incoming.

Table of Contents

1. The Module Action
2. From $\otimes$-MIKE to MIKE
3. MIKE as a NIKE

Reminder: Elliptic curves

Supersingular elliptic curves over $\mathbb{F}_{p^2}$:

- Variety of dim 1
- Abelian group structure
- *Supersingular* if $|E(\mathbb{F}_{p^2})| = (p+1)^2$

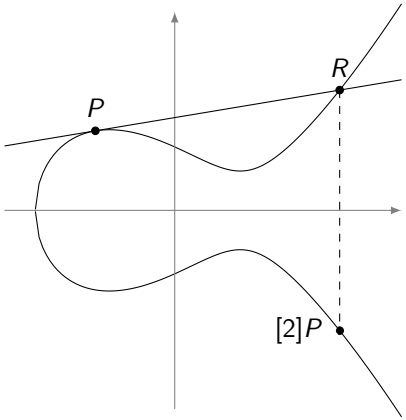


Figure: Elliptic curve

Reminder: Abelian Varieties

Maximal abelian varieties over $\mathbb{F}_{p^2}$:

- Variety of dim g
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- *Maximal* if $|A(\mathbb{F}_{p^2})| = (p+1)^{2g}$

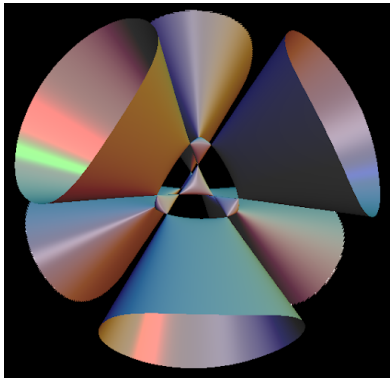


Figure: Abelian variety of dimension 2

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- *Isogenies*

$$\phi : A \longrightarrow B \cong A/\ker(\phi)$$

  - Characterised by their kernel.

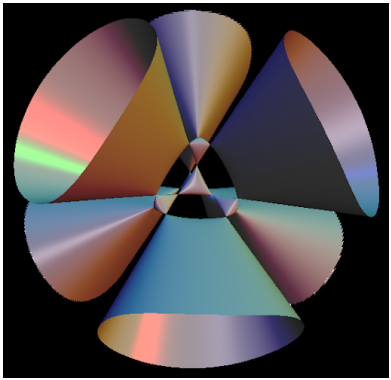


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- $\mathrm{End}(A)$ are orders of $\mathbb{Q}$-algebras.

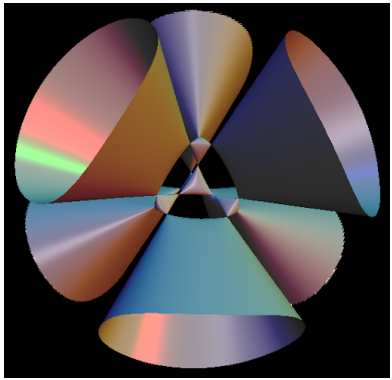


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Oriented Abelian Varieties

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R-Modules

Let M be a good[a] R-module with a torsion free s.e.s

$$R^n \xrightarrow{L_M \cdot -} R^m \longrightarrow M \longrightarrow 0$$

with $L_M \in \mathrm{Mat}_{m,n}(R)$ and $\mathrm{rk}(M) := m - n$.

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- $\mathrm{Hom}_{\mathcal{AV}_R}(A, B)$ are R-modules!

Module action

Let M a R-module of rank r and A a R-oriented abelian variety

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The **module action** $M \star_R A$ is the unique object in $\mathcal{AV}_R$ such that $\forall X \in \mathcal{AV}_R$

$$Hom_{\mathcal{AV}_R}(X, M \star_R A) \cong Hom_R\big(M, Hom_{\mathcal{AV}_R}(X, A)\big)$$

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Consequences

- By 2): the module action is (multiplicatively) graded

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$$\begin{aligned} Hom_{\mathcal{AV}_R}(X, M_2 \star_R (M_1 \star_R A)) &\cong Hom_R\big(M_2, Hom_{\mathcal{AV}_R}(X, (M_1 \star_R A))\big) \\ &\cong Hom_R\Big(M_2, Hom_R\big(M_1, Hom_{\mathcal{AV}_R}(X, A)\big)\Big) \\ &\cong Hom_R\Big(M_2 \otimes_R M_1, Hom_{\mathcal{AV}_R}(X, A)\Big) \\ &\cong Hom_{\mathcal{AV}_R}(X, (M_2 \otimes_R M_1) \star_R A) \end{aligned}$$

And Action!

Theorem

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$$\begin{array}{ccc} R\text{-Modules} & \xrightarrow{-\star_R E_0} & \mathcal{AV}_R \\ M & \longrightarrow & M \star_R E_0 \end{array}$$

is an symmetric anti-equivalence of category.

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- Generalisation of the isogeny group action.

The Cryptographic Module Action

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Let E_0 s.t. $R \cong \mathrm{End}_{\mathcal{AV}_R}(E_0)$.

$$\begin{array}{ccc} R\text{-Hermitian Modules} & \xrightarrow{-\star_R(E_0,\lambda)} & R\text{-oriented PPAVs} \\ (M, H_M) & \longrightarrow & (M, H_M) \star_R (E_0, \lambda) \end{array}$$

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This is our MIKE's cryptographic monoid action!

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$\otimes$-MIKE

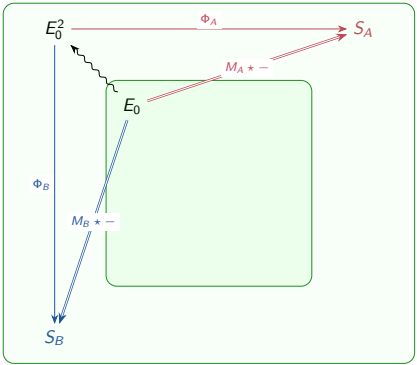
E_0

$p = c2^f - 1$, $R := \mathbb{Z}[\omega]$ with $\omega = \frac{1+\sqrt{-p}}{2}$.

$\otimes$-MIKE

Alice	$E_0 : y^2 = x^3 - x$	**Bob**
M_A of rank 2		M_B of rank 2

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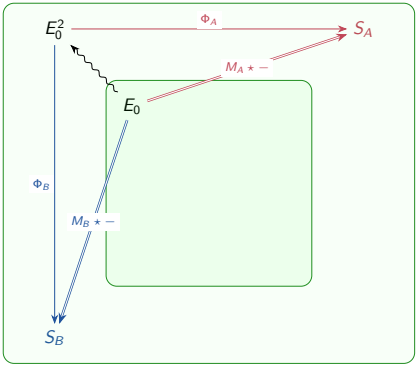


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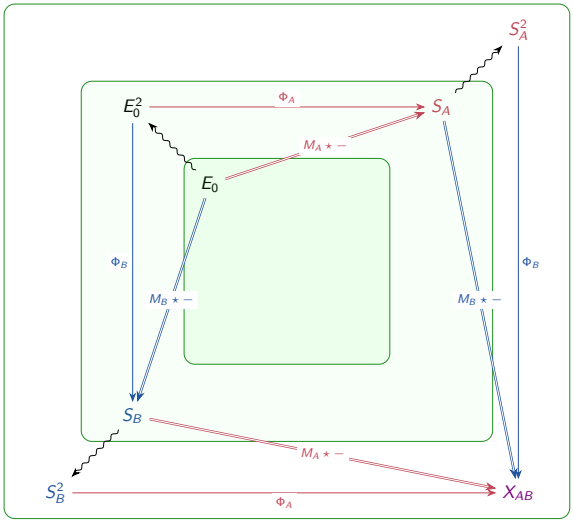
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← S_B

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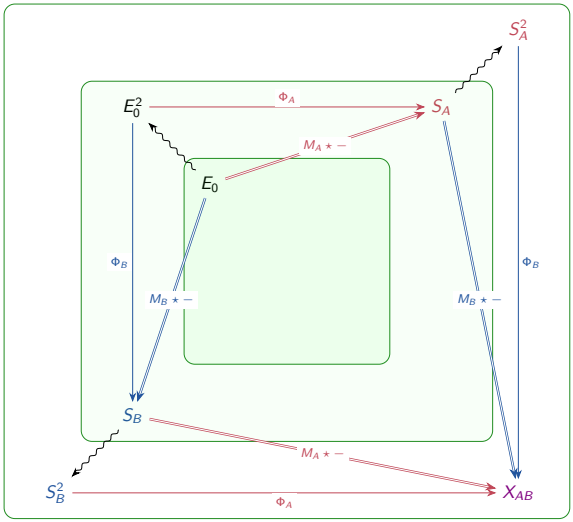


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Let $\mathcal{B}_{p,\infty}$ be the quaternion algebra, $\mathcal{O}_0 := \mathbb{Z}[\omega] + \mathbf{i}\mathbb{Z}[\omega]$.

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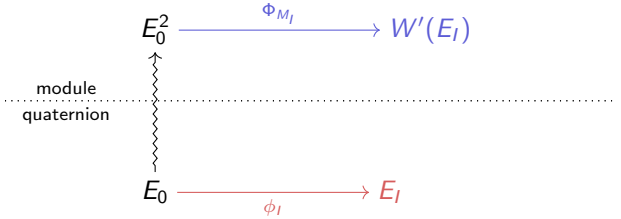
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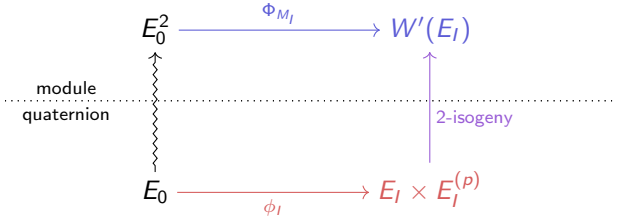
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- Smaller public keys (1 j-invariant over $\mathbb{F}_{p^2}$).
- Public key verification is free. (easy to check supersingularity).

MIKE's miraculous construction

MIKE's modules are of the form $I_x = \mathcal{O}_0\langle 2^e, (\omega - 1)(1 - x\mathbf{i})\rangle$ *with* $x = 3 \mod 8$

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MIKE's motto: A good quaternion is a non-implemented quaternion!

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MIKE's fourfolds

- On $(E \times E^{(p)})^2$

$$[\mathbf{i}] : (P_1, P_2, P_3, P_4) \longrightarrow (P_3, -P_4, -P_1, P_2)$$

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▶ All fourfolds in MIKE have $\mathbf{i} \in \mathrm{Aut}(X)$.

Theorem

MIKE's abelian fourfolds are Weil type-$(2,2)$ fourfolds!

▶ They are very special.

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- We represent fourfold in the theta model

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i-compatible theta model

For X a Weil fourfold and θ^X s.t. $K_T = \ker(1 + \mathbf{i})$:

$$\forall P \in X, \quad \theta_j^X\big([\mathbf{i}](P)\big) = (-1)^{\delta_j} \cdot \theta_j^X(P)$$

with $\delta_j = 1$ for $j = 5, 7, 10, 11, 13, 14$ and 0 otherwise.

Weil type invariants

$$\theta_j^X(0_X) = (a_0 : a_1 : a_2 : a_3 : a_4 : 0 : a_6 : 0 : a_8 : a_9 : 0 : 0 : a_{12} : 0 : 0 : a_{15})$$

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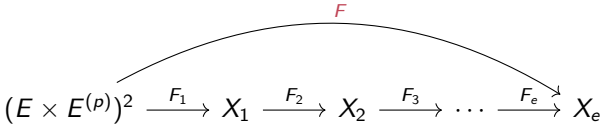
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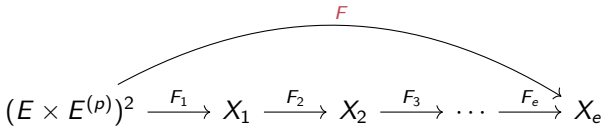
- Non trivial!

Computing isogenies between Weil varieties



with all X_i of Weil type.

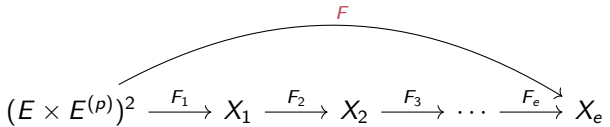
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Theorem

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$$\forall P \in X, \quad \theta_j^X\big([\mathbf{i}](P)\big) = \theta^X_{\sigma \circ j}(P)$$

with $\sigma = (03)(12)$.

Fast isogenies between Weil varieties

Isogeny theorem

Computing HD isogenies

$\Longleftrightarrow$

interpolating $\theta^X(0_X)$

$\Longleftrightarrow$

Finding spanning trees in graphs.

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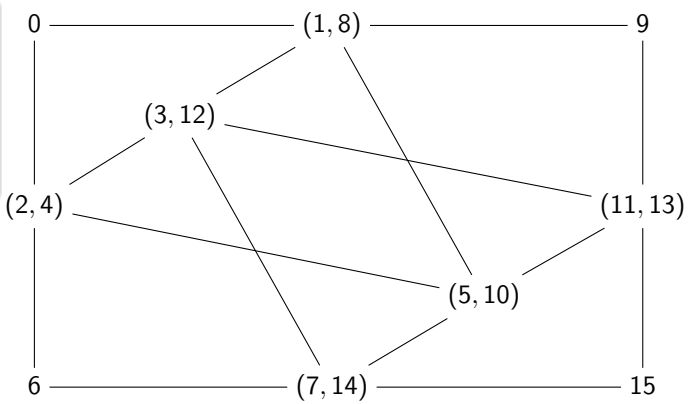
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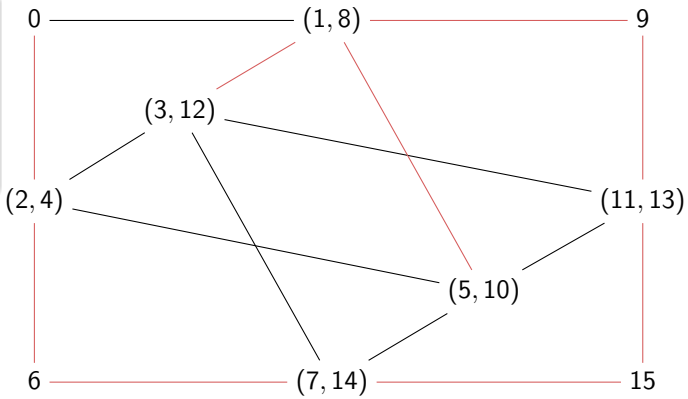
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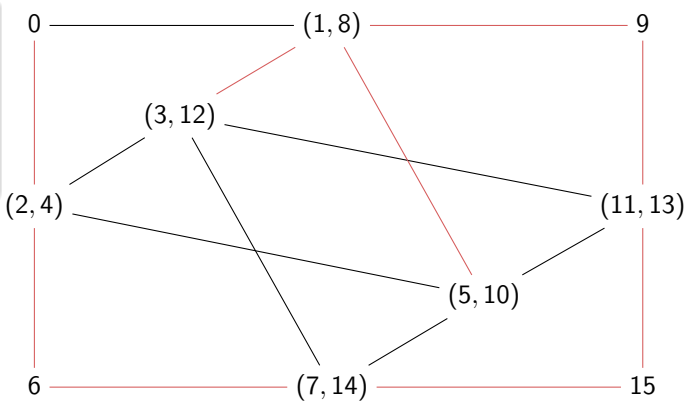
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By rabbit interpolation

MIKE's 4D isogenies are $\simeq$ 2x faster than generic !

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1. The Module Action
2. From $\otimes$-MIKE to MIKE
3. MIKE as a NIKE

MIKE

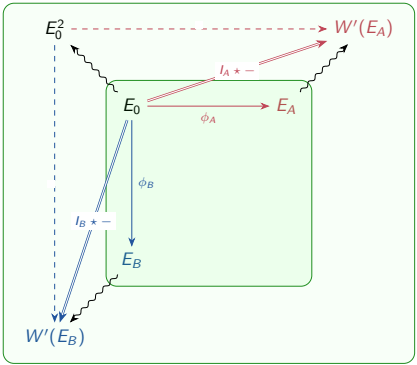
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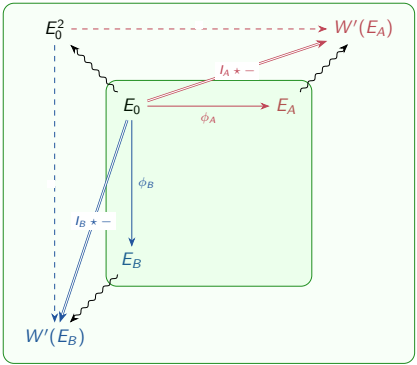


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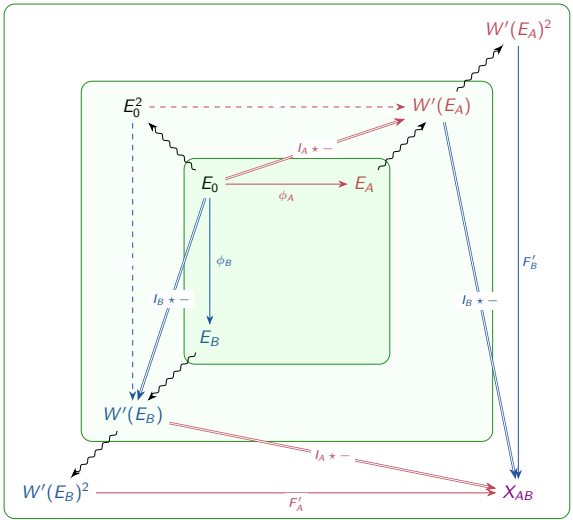


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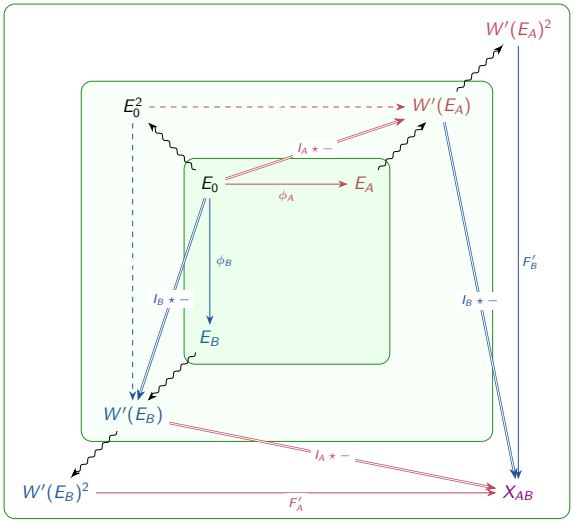


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Unlike Diffie-Hellman:

- No group structure (e.g. No collision[1])

$$X_{AB} \cong X_{A'B'} \iff E_{A'} \cong E_A \text{ or } E_A^{(p)} \text{ and } E_{B'} \cong E_B \text{ or } E_B^{(p)}$$

[1] conjectured and experimentally verified

MIKE Security

MIKE is a post-quantum NIKE

Like Diffie-Hellman:

- MIKE DLog: $E_A \xrightarrow{\text{compute}} \phi_A$

  MIKE DLog = IsogenyProblem

- MIKE CDH: $E_A, E_B \xrightarrow{\text{compute}} X_{AB}$
  - MIKE is *passively secure* if CDH is hard.
  - MIKE is *actively secure* in QROM if gapCDH is hard.

Unlike Diffie-Hellman:

- No group structure (e.g. No collision[1])

$$X_{AB} \cong X_{A'B'} \iff E_{A'} \cong E_A \text{ or } E_A^{(p)} \text{ and } E_{B'} \cong E_B \text{ or } E_B^{(p)}$$

- No Kuperberg!

[1] conjectured and experimentally verified

MIKE in the AIM+

Theorem

In the *generalised Algebraic Isogeny Model* (AIM+):

$$\text{MIKE CDH}^a \iff \text{EndRing}$$

[a] with isogenies degree $\simeq p$

$E_i \in \{E_0, E_A, E_B\}$

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E_0 E_A

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E_A

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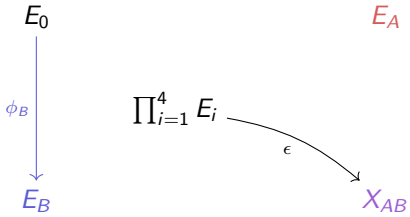
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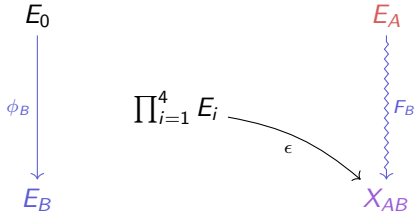
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$$\prod^{4} E_i \xrightarrow{\epsilon} X_{AB} \xrightarrow{\widetilde{F_B}} {E_A}^4$$

$$\Downarrow \text{w.p. } 1/2$$

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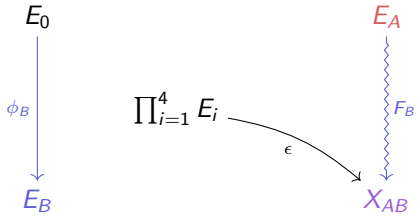
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- Solution of the 7th Isogeny Problem [CDFG+26]!

$E_i \in \{E_0, E_A, E_B\}$

MIKE Performance in C

		Parameters / Sizes				x86 `ref`		x86 `asm`		ARM `asm`	
		p	$\log_2(p)$	e	pk B	KG MCy	SK MCy	KG MCy	SK MCy	KG MCy	SK MCy
Fast	I	$633 \cdot 2^{308} - 1$	318	228	80	3.75	27.3	2.15	16.0	1.09	7.80
	III	$593 \cdot 2^{474} - 1$	484	340	122	9.22	69.2	6.96	49.3	3.25	24.3
	V	$317 \cdot 2^{628} - 1$	637	452	160	21.9	146	13.7	94.2	6.36	47.6
Slow	I	$117 \cdot 2^{374} - 1$	381	372	96	8.24	61.7	4.90	34.7	2.38	16.7
	III	$77 \cdot 2^{566} - 1$	573	564	144	24.0	161	15.0	103	6.89	50.7
	V	$41 \cdot 2^{758} - 1$	764	756	192	51.8	340	36.8	230	15.5	118

Figure: Benchmarks of C implementation of MIKE[2]

[2] Timing measured on AMD Ryzen 7 PRO 7840U@3.3GHz & ARM M4 Pro@4.5GHz.
If you wonder why 2 sets of parameters, blame Benjamin Wesolowski [Wes26]

MIKE Performance in C

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Figure: Benchmarks of C implementation of MIKE[2]

Implementation is:

- Constant time!
- Simple! ($\sim$ 10k LoC)
  - even gluing!

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The end (of this introduction)

This was an otherview of MIKE

Tons of things I skipped.

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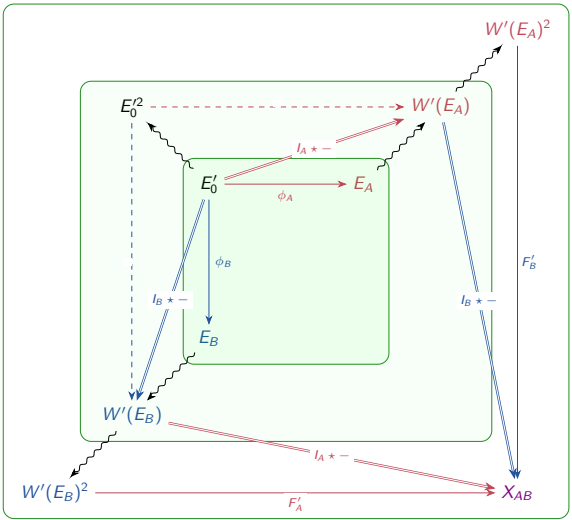
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- 100 pages of appendices waiting for you!

https://mike.isogeni.es/



References I

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